LIM-1.E.1 It may be necessary or helpful to rearrange expressions into equivalent forms before evaluating limits.

LIM-1.E.2 The limit of a function may be found by using the squeeze theorem.

1-6 & 1-7 Limits Using Algebraic Manipulation

Get Ready...

Evaluate the following limits, if possible.

A) $\lim_{x\to 3}(2x^2-3) = 18-3 = 15$

B) $\lim_{x\to -2} 5 = 5$

C) $\lim_{x\to 1}\frac{x^2+x+2}{x+1} = \frac{1+1+2}{1+1} = \frac{4}{2} = 2$

D) $\lim_{x\to \frac{\pi}{6}}\sin(x) = \sin\frac{\pi}{6} = \frac{1}{2}$

Evaluating Limits Analytically

While the concept of a limit works for both continuous and discontinuous functions, the AP exam will target functions that are not continuous. However, if working with a continuous function, don't overcomplicate the idea of a limit. **With a continuous function, the limit is simply the same value as the actual point on the graph.**

What if the function is not continuous(otherwise known as discontinuous)?

In the cases above, nothing of great interest (or new) happened when we attempted to find the limit. However, that is not always the case.

Example 1: Consider the following limit, $\lim_{x \to 4} \frac{x^2-3x-4}{x^2-9x+20}$. What do we get when we attempt to "plug in" $x = 4$?

$$\lim_{x \to 4} \frac{x^2-3x-4}{x^2-9x+20} = \frac{16-12-4}{16-36+20} = \frac{0}{0}$$

Indeterminate Forms

While there are many different types of indeterminate forms, we will only deal with two in AP Calculus:

$$\frac{0}{0} \ \ and \ \frac{\infty}{\infty}$$

When we get an indeterminate form, there are several techniques we can use to evaluate the limit.

Technique 1: Factoring (Use this technique when the limit is a fraction of polynomials)

Evaluate the following limits, if possible.

Example 2: $\lim_{x\to 4} \frac{x^2-3x-4}{x^2-9x+20} = \lim_{x\to 4} \frac{(x-4)(x+1)}{(x-4)(x-5)} = \lim_{x\to 4} \frac{x+1}{x-5}$

$= \frac{4+1}{4-5} = \frac{5}{-1} = -5$

Example 3: $\lim_{x \to 0} \frac{4x^3 - 5x^2}{6x^2 + 8x^5} =$

$$\lim_{x \to 0} \frac{4x^3 - 5x^2}{6x^2 + 8x^5} = \lim_{x \to 0} \frac{x^2(4x-5)}{2x^2(3+4x^3)}$$

$$= \lim_{x \to 0} \frac{4x-5}{2(3+4x^3)} = \frac{-5}{2(3)} = \frac{-5}{6}$$

Example 4: (From the 2013 AP exam)

$$\lim_{x\to 2}\frac{x^2+x-6}{x^2-4} = \lim_{x\to 2}\frac{(x+3)\cancel{(x-2)}}{(x+2)\cancel{(x-2)}} = \lim_{x\to 2}\frac{x+3}{x+2} = \frac{5}{4}$$

(a) $-\frac{1}{4}$ (b) 0 (c) 1 (d) $\frac{5}{4}$ (e) nonextistent

Technique 2: Rationalizing a numerator (or denominator) (Use this technique when dealing with radicals)

Evaluate the following limits, if possible.

Example 5: $\lim_{x\to 0} \frac{\sqrt{x+1}-1}{x} =$

$$\lim_{x\to 0} \frac{\sqrt{x+1}-1}{x} \cdot \frac{\sqrt{x+1}+1}{\sqrt{x+1}+1}$$

$$= \lim_{x\to 0} \frac{x+1-1}{x(\sqrt{x+1}+1)}$$

$$= \lim_{x\to 0} \frac{x}{x(\sqrt{x+1}+1)}$$

$$= \lim_{x\to 0} \frac{1}{\sqrt{x+1}+1} = \frac{1}{1+1} = \frac{1}{2}$$

Example 6: $\lim_{x\to 0} \frac{\sqrt{x+5}-\sqrt{5}}{x} =$

$$\lim_{x\to 0} \frac{\sqrt{x+5}-\sqrt{5}}{x} \cdot \frac{\sqrt{x+5}+\sqrt{5}}{\sqrt{x+5}+\sqrt{5}}$$

$$= \lim_{x\to 0} \frac{x+5-5}{x\left(\sqrt{x+5}+\sqrt{5}\right)}$$

$$= \lim_{x\to 0} \frac{x}{x\left(\sqrt{x+5}+\sqrt{5}\right)}$$

$$= \lim_{x\to 0} \frac{1}{\sqrt{x+5}+\sqrt{5}} = \frac{1}{\sqrt{5}+\sqrt{5}} = \frac{1}{2\sqrt{5}}$$

Technique 3: Multiplying by a "Fancy Form Of One" (Use this technique for Complex Fractions)

Evaluate the following limits, if possible.

Example 7: $\lim_{x\to 0} \frac{x}{\frac{1}{x-4}+\frac{1}{4}}$

$$\lim_{x\to 0} \frac{x}{\frac{1}{x-4}+\frac{1}{4}} \cdot \frac{(x-4)4}{(x-4)4}$$

$$= \lim_{x\to 0} \frac{4x(x-4)}{4+x-4}$$

$$= \lim_{x\to 0} \frac{4x(x-4)}{x}$$

$$= \lim_{x\to 0} 4(x-4) = 4(-4) = -16$$

$$\frac{1}{\cancel{x-4}} \cdot \cancel{(x-4)}4$$

$$\frac{1}{\cancel{4}}(x-4)\cancel{(4)}$$

Example 8: $\lim_{x\to 0} \frac{\frac{1}{(x+3)^2} - \frac{1}{9}}{x} \cdot \frac{(x+3)^2(9)}{(x+3)^2(9)}$

$$= \lim_{x\to 0} \frac{9-(x+3)^2}{9x(x+3)^2}$$

$$= \lim_{x\to 0} \frac{9+(-x^2-6x-9)}{9x(x+3)^2}$$

$$= \lim_{x\to 0} \frac{-x^2-6x}{9x(x+3)^2}$$

$$= \lim_{x\to 0} \frac{-x(x+6)}{9x(x+3)^2}$$

$$= \lim_{x\to 0} \frac{-(x+6)}{9(x+3)^2} = \frac{-6}{81} = \frac{-2}{27}$$

Technique 4: Simplifying the expression

Example 9: Evaluate: $\lim_{h\to 0}\frac{(3+h)^2-9}{h}$

$(3+h)(3+h)$

$$= \lim_{h\to 0} \frac{9+6h+h^2-9}{h}$$

$$= \lim_{h\to 0} \frac{6h+h^2}{h}$$

$$= \lim_{h\to 0} (6+h) = 6+0 = 6$$

Special Case:

Absolute Value expressions

Limits that involve absolute value expressions have to be converted to a piecewise function to calculate the limit. To do this, you need to remember the definition of the absolute value function:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

or in more general terms: $|\sim\sim\sim| = \begin{cases} (\sim\sim\sim) & \text{if } (\sim\sim\sim) \geq 0 \\ -(\sim\sim\sim) & \text{if } (\sim\sim\sim) < 0 \end{cases}$

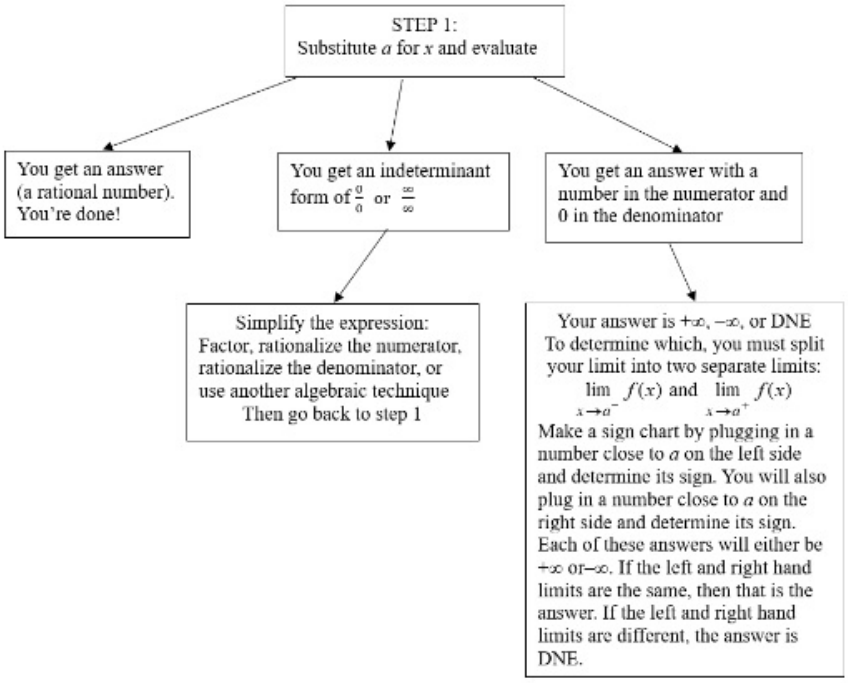
Example 10: Evaluate $\lim_{x\to 3} \boxed{\frac{x-3}{|x-3|}}$ $\Rightarrow$ DNE

$$\lim_{x\to 3^-} \frac{x-3}{|x-3|} = \lim_{x\to 3^-} \frac{x-3}{-(x-3)} = \lim_{x\to 3^-} (-1) = -1$$

$$\lim_{x\to 3^+} \frac{x-3}{|x-3|} = \lim_{x\to 3^+} \frac{x-3}{x-3} = \lim_{x\to 3^+} 1 = 1$$

$$\frac{x-3}{|x-3|} = \begin{cases} \frac{x-3}{x-3} & \text{if } x-3 \geq 0 \quad x \geq 3 \\ \frac{x-3}{-(x-3)} & \text{if } x-3 < 0 \quad x < 3 \end{cases}$$

How to find an algebraic limit of $\lim_{x \to a} f(x)$



Example 11: Evaluate $\lim_{x \to 2} \frac{2x+5}{x-2}$ $= DNE$

$\frac{9}{0}$

$-\infty \quad | \quad +\infty$

2

$1.9 \qquad 2.1$

$\frac{+}{-} \qquad \frac{+}{+}$